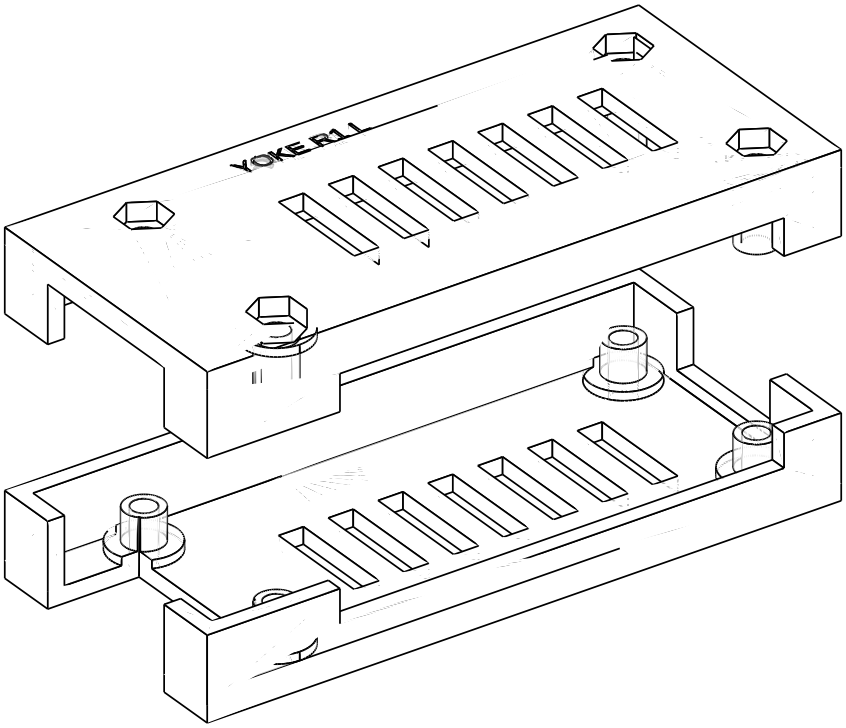
Inventory and orientation

Print: one base marked R1 B and one lid marked R1 L; both outside-face down.
Supply: one ZERO 3E V1.2 board, four M2.5 x 20 pan-head screws,
four M2.5 hex nuts (AF5, 2mm thick), driver and3mm hand drill bit.
20mm is measured beneath the screw head. Never substitute a longer screw.
The case body is 76.4 x 34.8 x 20; screw heads project about1.7 below.
Test equipment: scrap wood, caliper, padded mat,0-10 N spring gauge/scale,
actual Ethernet plus host and OTG/power USB-C cables, and a powered
test host with CPU temperature readout and a30-minute timer.

R1 L: flat exterior above; nut wells outside

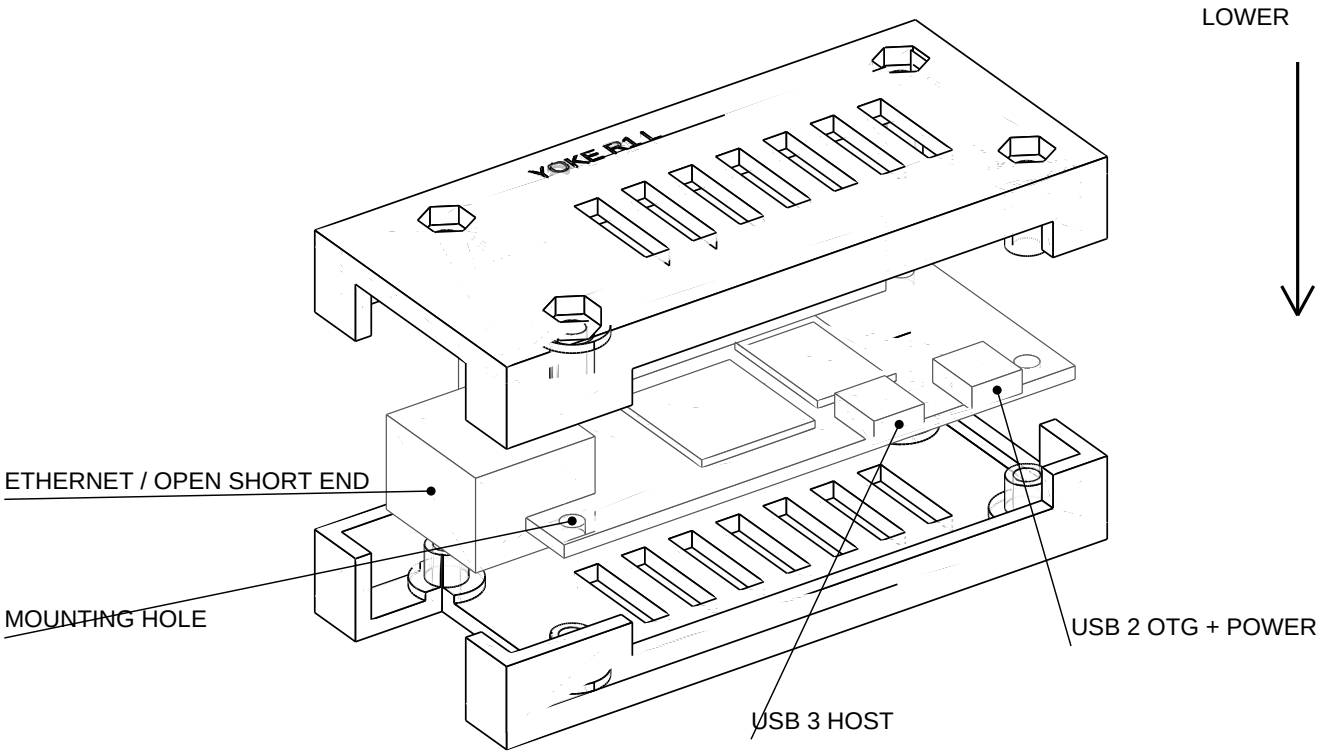


R1 B: floor below; four raised mounting posts

1. Let both parts cool before lifting them from the print surface.
2. With lid alone on scrap wood, turn a3mm bit BY HAND through its
   four0.20mm membranes and lightly ream lid bores. Remove all swarf.
3. Require lid bores3.0-3.2mm and four intact flat nut seats; no torn ledges.
4. Check each nut seats flat. A tight nut, damaged seat or warped part fails.

Keep the board unpowered throughout assembly. No glue or inserts are used.
Do not press plastic against IC packages, GPIO pins or connector shells.

Lower the board, then the lid



Board context above is simplified; official STEP and raw mesh are bound.

1. Face the Ethernet connector toward the open short end of R1 B.
2. Lower the board component-side up onto all four posts.
   Each post supports only the bare corner mounting region.
3. Align R1 L and lower it vertically. Keep both USB-C openings clear.
   The lid posts meet the upper mounting pads; the chips retain clearance.
4. While upright, drop one nut into each of the four exterior lid wells.
   Keep the enclosure upright so loose nuts cannot fall out.

If the board rocks, a component touches, or the lid needs force: STOP.

The nominal0.15 seam gap lets the corner pads clamp before wall contact.

One mounting joint / section through screw axis

Section Y=3.599942 through mount1; exact final meshes; enlarged3.53:1.

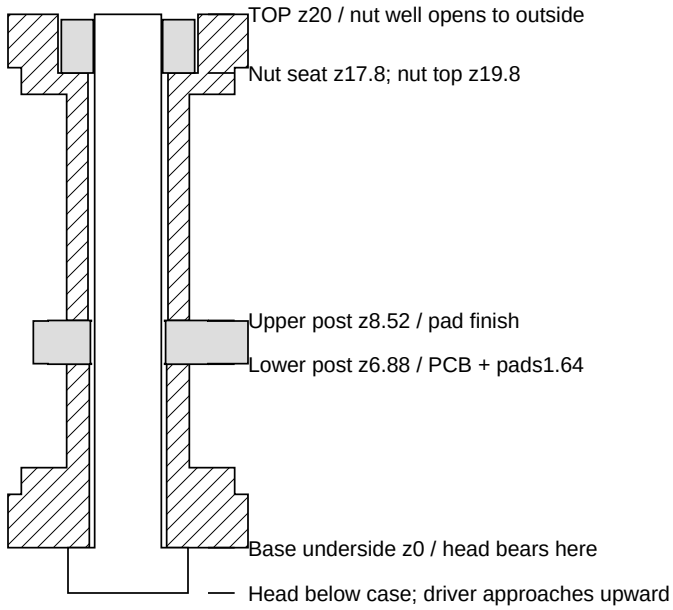
Shown after0.20mm webs are cleared and lid bores hand-reamed to3mm.

White + hatch: printed plastic

Gray: PCB + pads1.64mm (laminate1.60)

White shaft: M2.5 x20 screw

Gray top block: AF5 nut



1. Hold upright above the table edge. Insert each screw from below.
2. Start all four threads before tightening; do not cross-thread.
3. Lightly snug alternating corners. The nut well prevents rotation.
4. Stop when the board is held; do not squeeze the seam fully closed.

No numerical tightening torque is qualified. Keep preload low.

Flat nut seat: z17.8, bore3mm. Clear membranes and lightly hand-ream lid bores.

Inspect the broad seat: no sag, crack, gouge or remaining swarf.

Stop for a spinning nut, whitening, cracking, PCB bow or force before seating.

Service: unplug power/cables, undo screws while upright, catch nuts,
lift the lid vertically, then lift the board. Inspect ledges before reassembly.

YOKE USB HOUSING R1 / 4 OF 4

First-article fit, use and service test

All observations below are NOT RUN until checked on the real article.

UNPOWERED FIT

1. Confirm the actual board says ZERO 3E V1.2. Count hardware and check20mm length.
2. Inspect every post, hole and nut ledge. Reject warp, strings or cracked seats.
3. Assemble without force. Require all four pads seated, no component contact,
   no visible board bow and no loose PCB. Check the seam is about0.05-0.25mm.

CABLE ACCESS AND RETENTION

4. Fully insert and release the actual Ethernet and both USB-C cables five times.
   Require latch access, complete engagement and no plastic levering a port.
   USB-C overmould assumption:12mm width x7mm height; actual cables must pass.
5. Over the padded mat, use the0-10 N spring gauge/scale to apply5 N
   in each cable direction for10s; support the case by hand.
   Require no case shift, cracking or changed connector seating.

POWERED THERMAL CHECK - AFTER THE ABOVE PASSES

6. Use the powered test host CPU readout and timer. Keep vents open at room
   temperature; run the intended USB load for30minutes.
   Record CPU temperature, ambient, time and interventions. Stop at75C,
   unexpected heating, or a lower manufacturer limit. Revise cooling on failure.

RECORD AND DECIDE

Record revision, filament, measured seam, pad contact, nut fit, cable models,
pull/cycle results and temperature. A print start proves none of these.
Stop for force, bowing, cracks, whitening, inaccessible latch or temperature limit.
Return failed fit to the CAD interface and repeat assembly review after edits.

LIMITS

The official STEP is nonmanifold: board Boolean checks are unavailable.
Corner triangle bounds exclude component surfaces from every post footprint.
Global nominal clearance: floor1.24mm below lowest component; roof1mm above
GPIO tips. Actual tolerance, strength and thermal performance require this test.
This prototype excludes a PoE HAT, added heatsink and ingress protection.